

Trusses drawn by the analysis program

PyNiteFEA 3.0.0 + native PyVista rendering

Every image comes from the model PyNite actually solved.

All connection-offset elements are included; no custom centerline drawing.

Each network is connected and independently reconstructed from its JSON data file.

Truss	Stage	Nodes	Elements	Components	Max vertical in
TE	complete	155	166	1	0.043
TE	roof_first	154	165	1	0.022
TW	complete	162	177	1	0.149
TW	roof_first	158	173	1	0.099
TN	complete	138	155	1	0.117
TN	roof_first	126	143	1	0.011
TS	complete	69	80	1	0.018
TS	roof_first	69	80	1	0.012
T1	complete	93	104	1	0.128
T1	roof_first	53	60	1	0.067

Displacements shown for completed storage service load or roof-first service load.

They include movement of the surrounding frame, not just truss deflection between bearings.

Whole-frame second-order screen: max partial member ratio 0.83; max floor-support movement 0.55 in.
Proposed corrections: join TE/TW upper chords at the slope transition; connect TN jamb base.

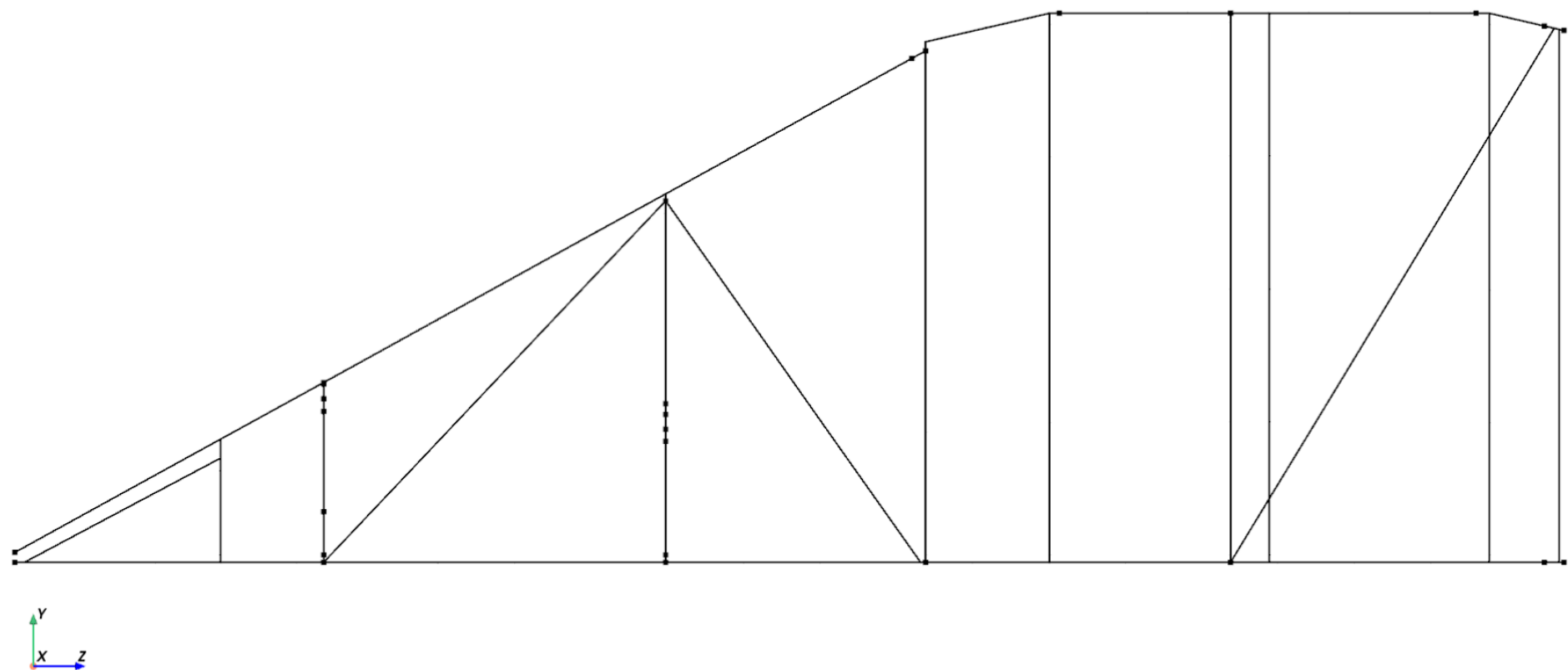
Whole-frame analysis was refreshed, then each truss independently solved and verified.

The three connections remain proposed details; their stiffness assumptions are explicit.

Boundary glyphs are imposed movements, not new ground supports.

PRELIMINARY: connection detailing, global iteration and capacity checks remain.

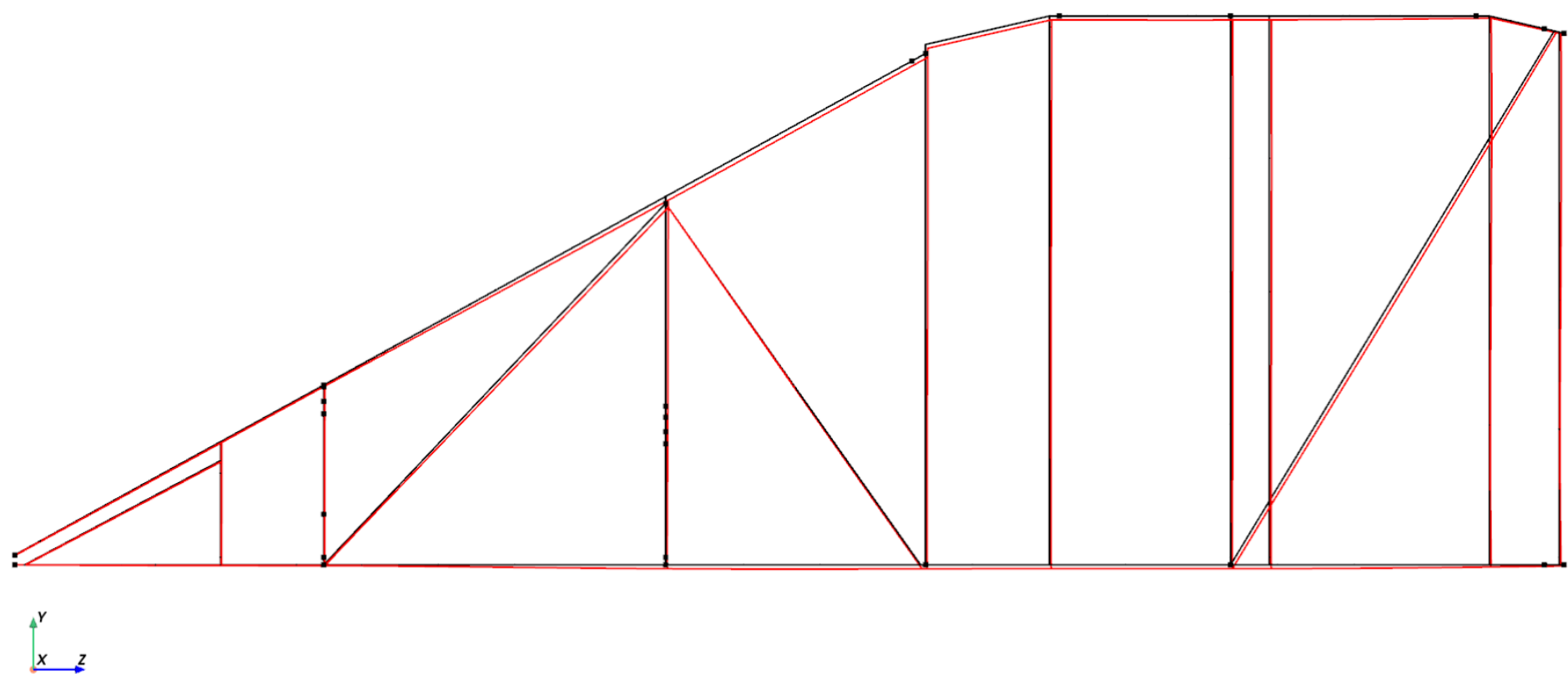
TE complete PyNite solved network



All physical members AND connection-offset elements shown. Nodes are real solver nodes.
Boundary glyphs represent imposed global-frame movements, not additional ground supports.

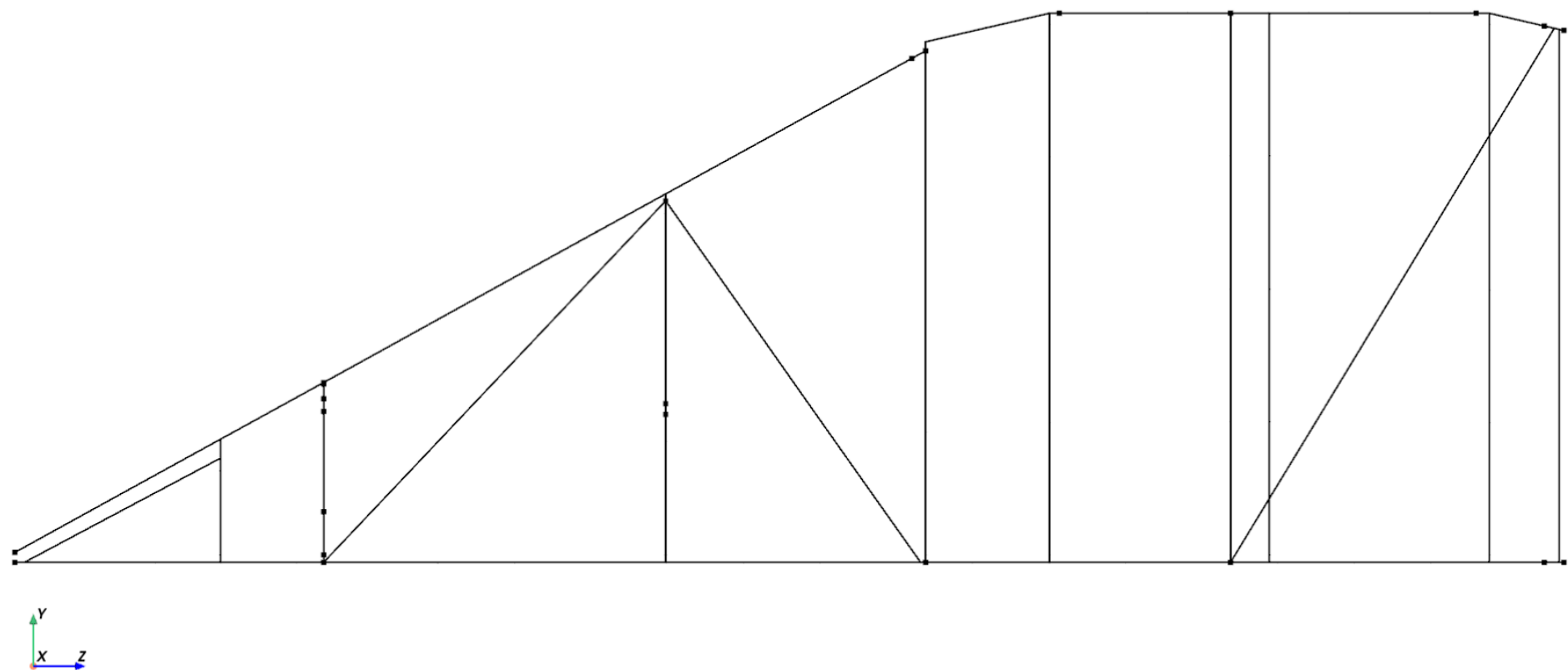
TE complete PyNite solved network

RED: displacement x20 service_patches



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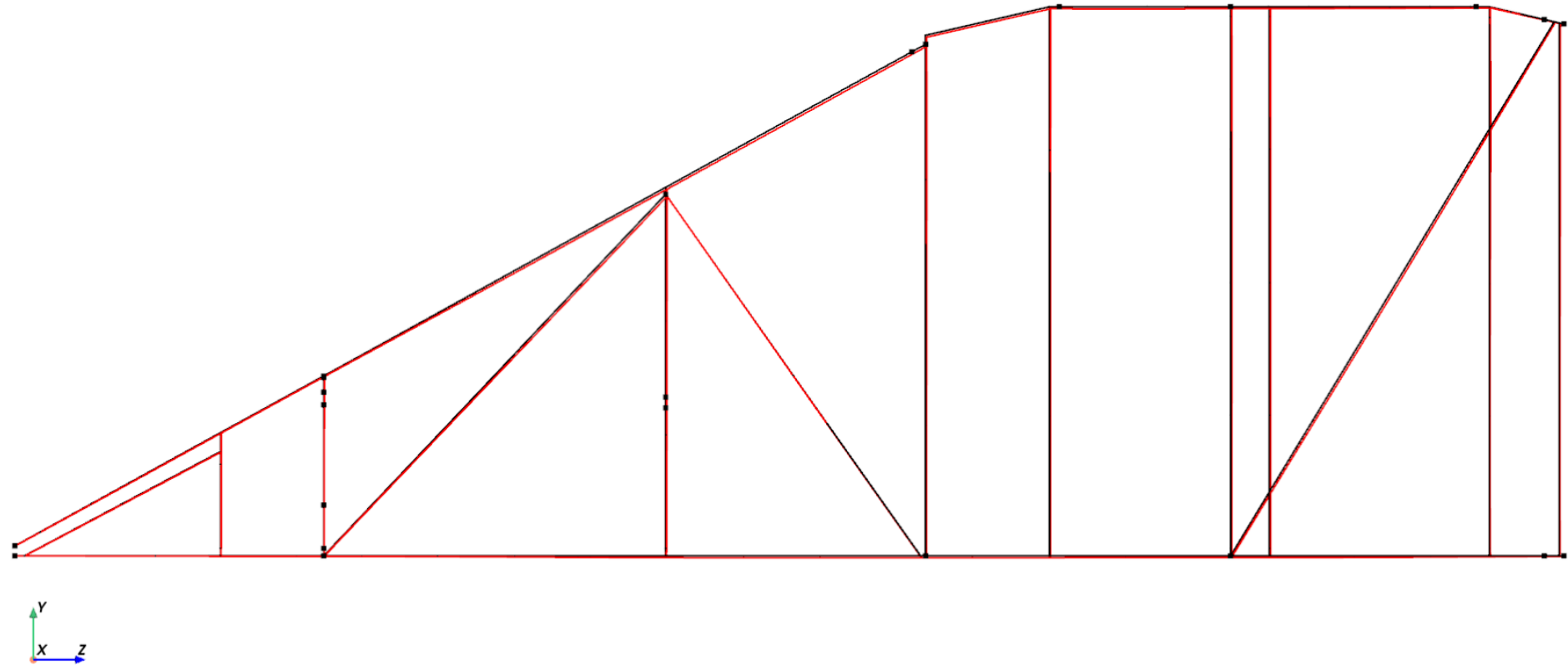
TE roof_first PyNite solved network



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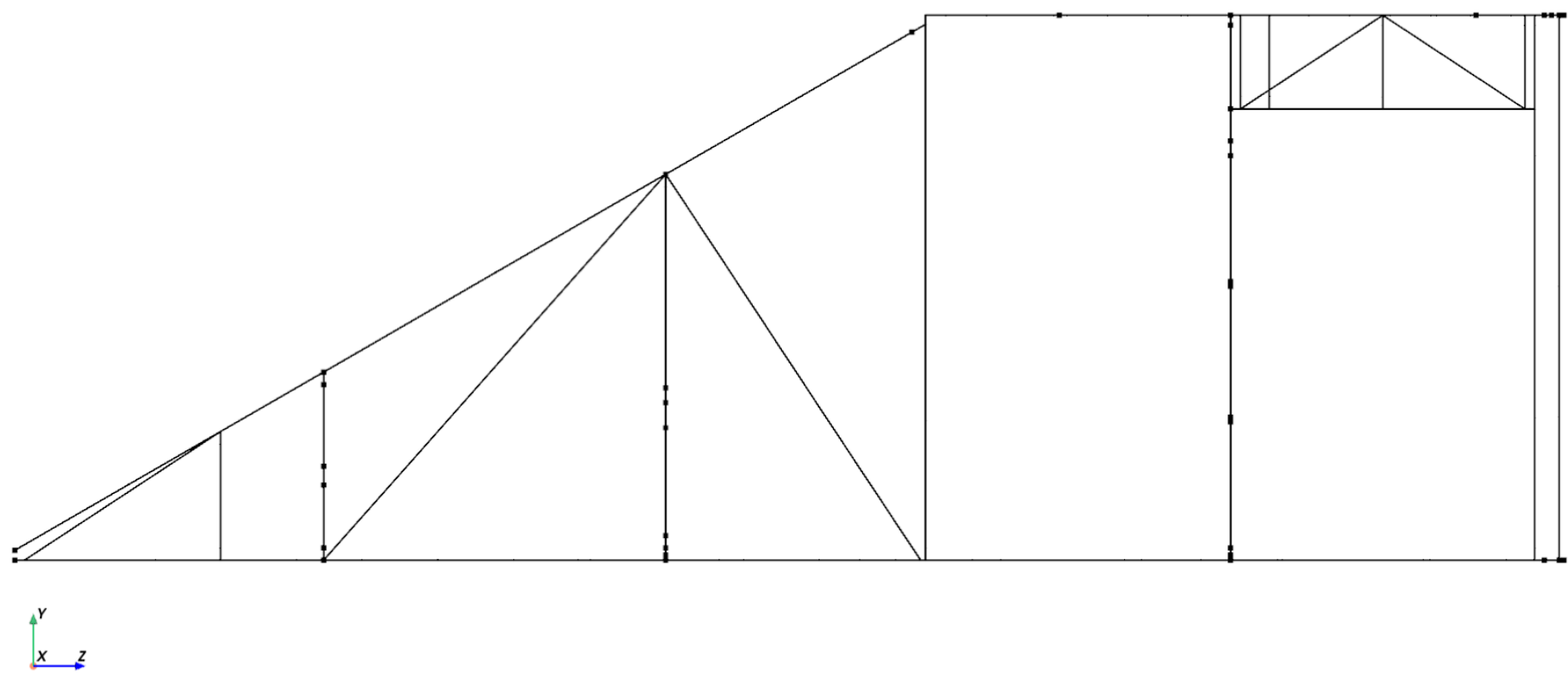
TE roof_first PyNite solved network

RED: displacement x20 roof_first_service



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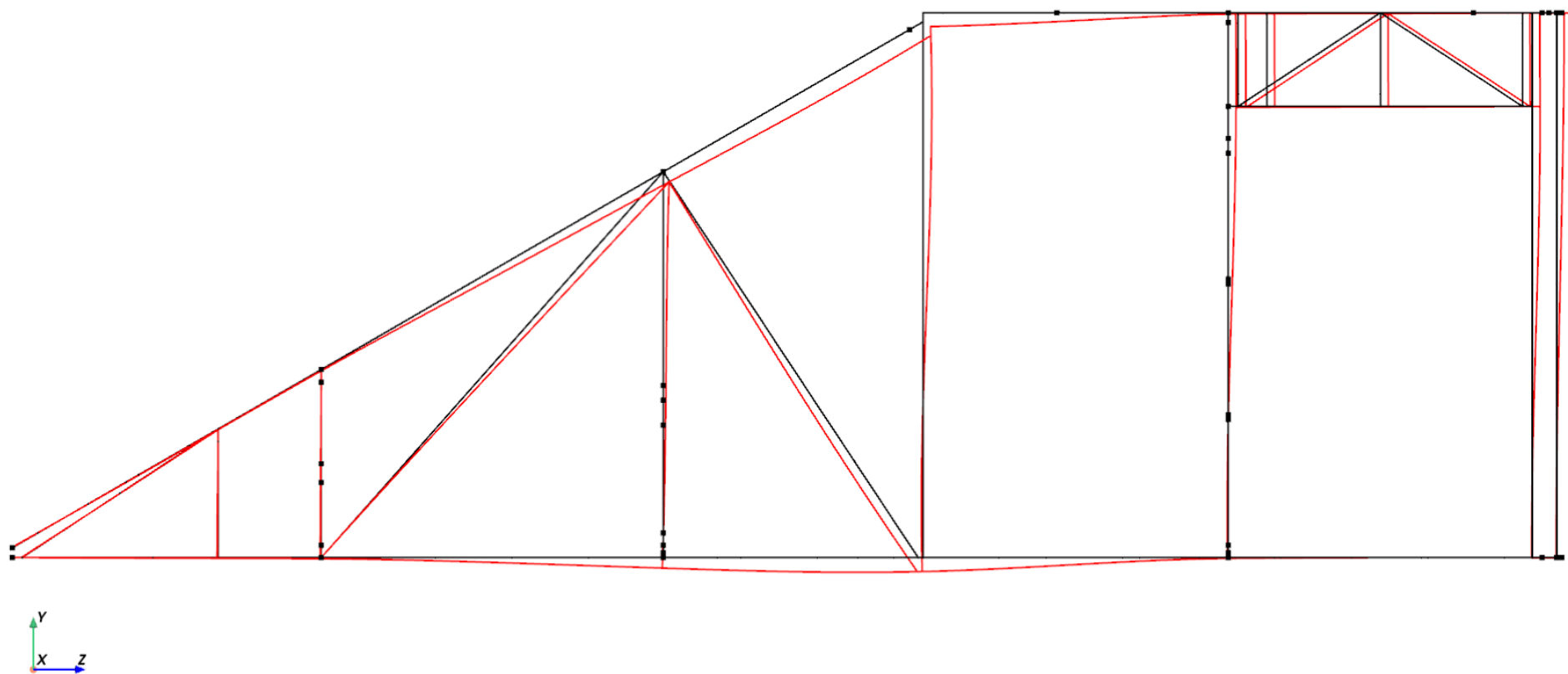
TW complete PyNite solved network



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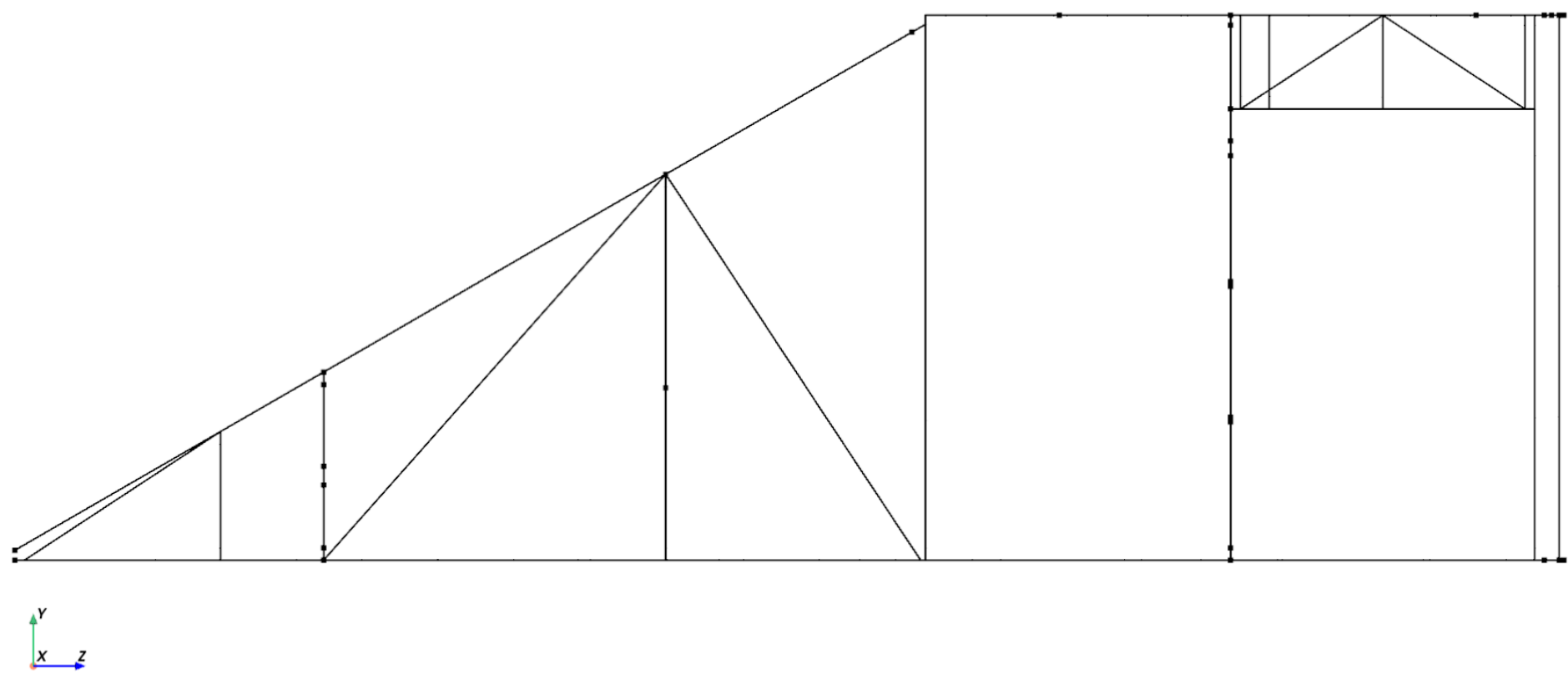
TW complete PyNite solved network

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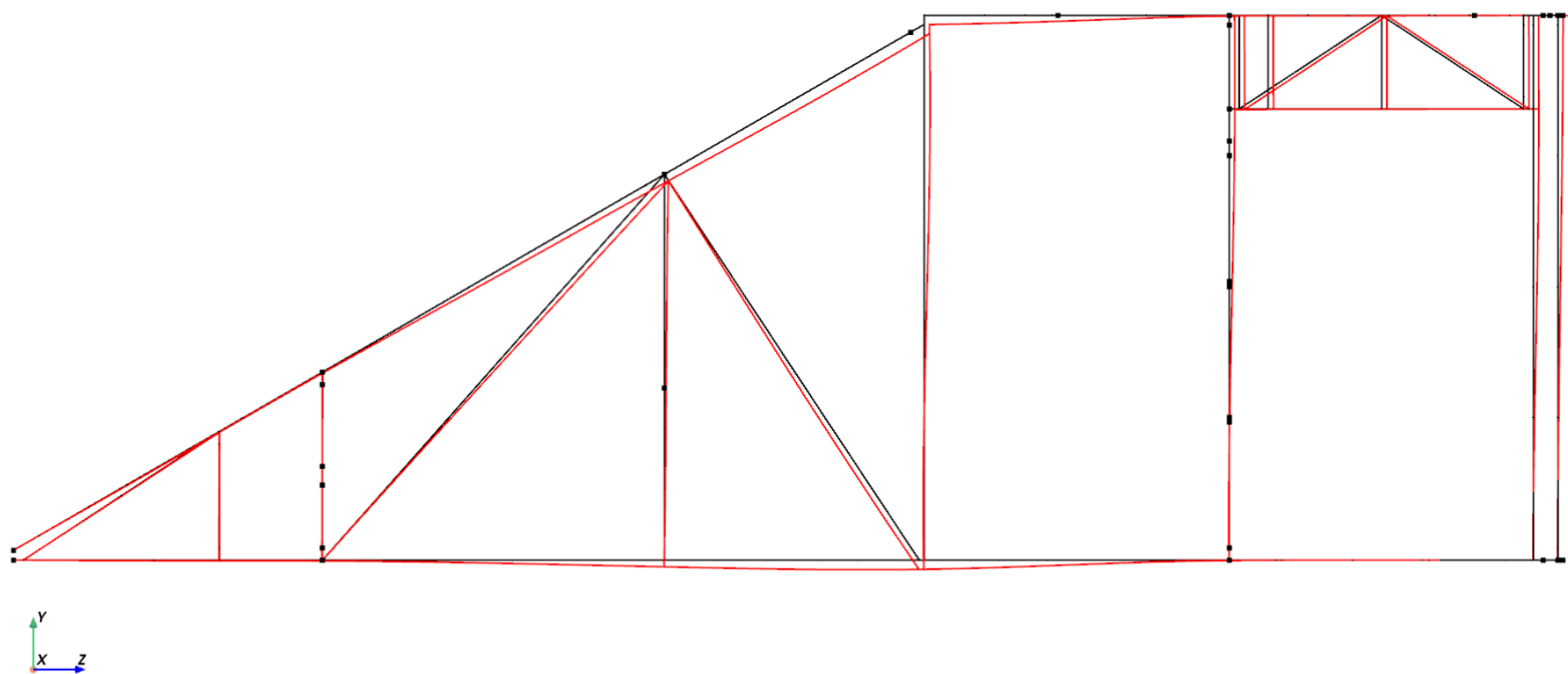
TW roof_first PyNite solved network



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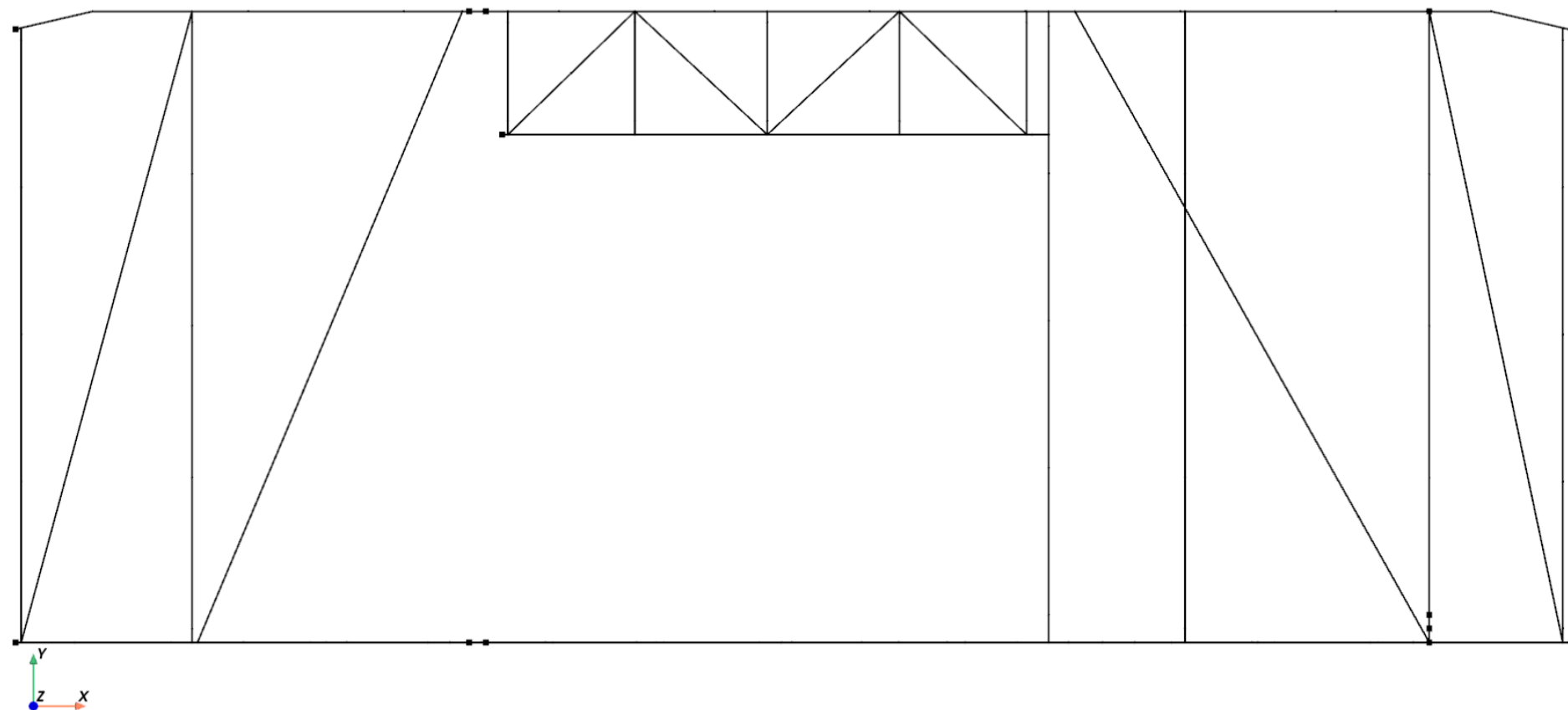
TW roof_first PyNite solved network

RED: displacement x20 roof_first_service



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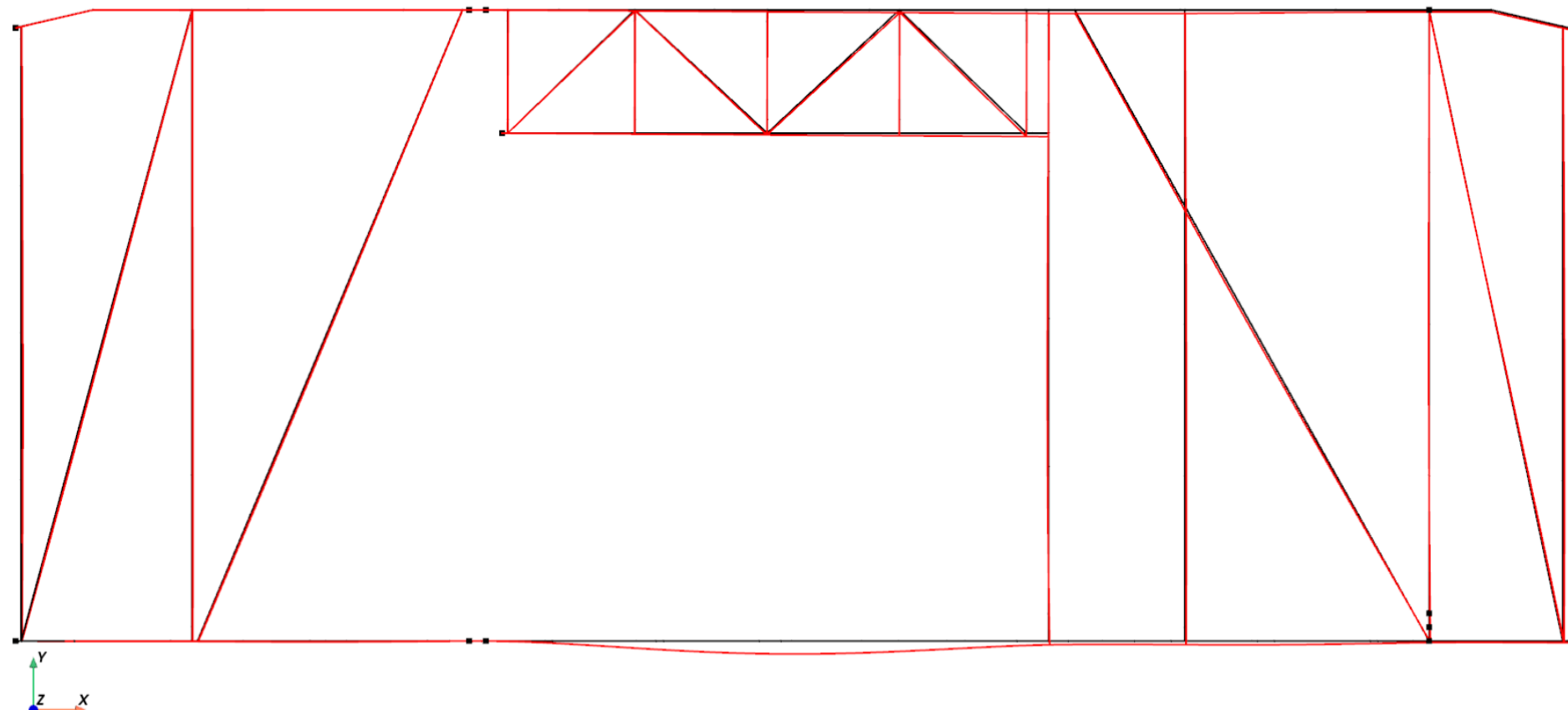
TN complete PyNite solved network



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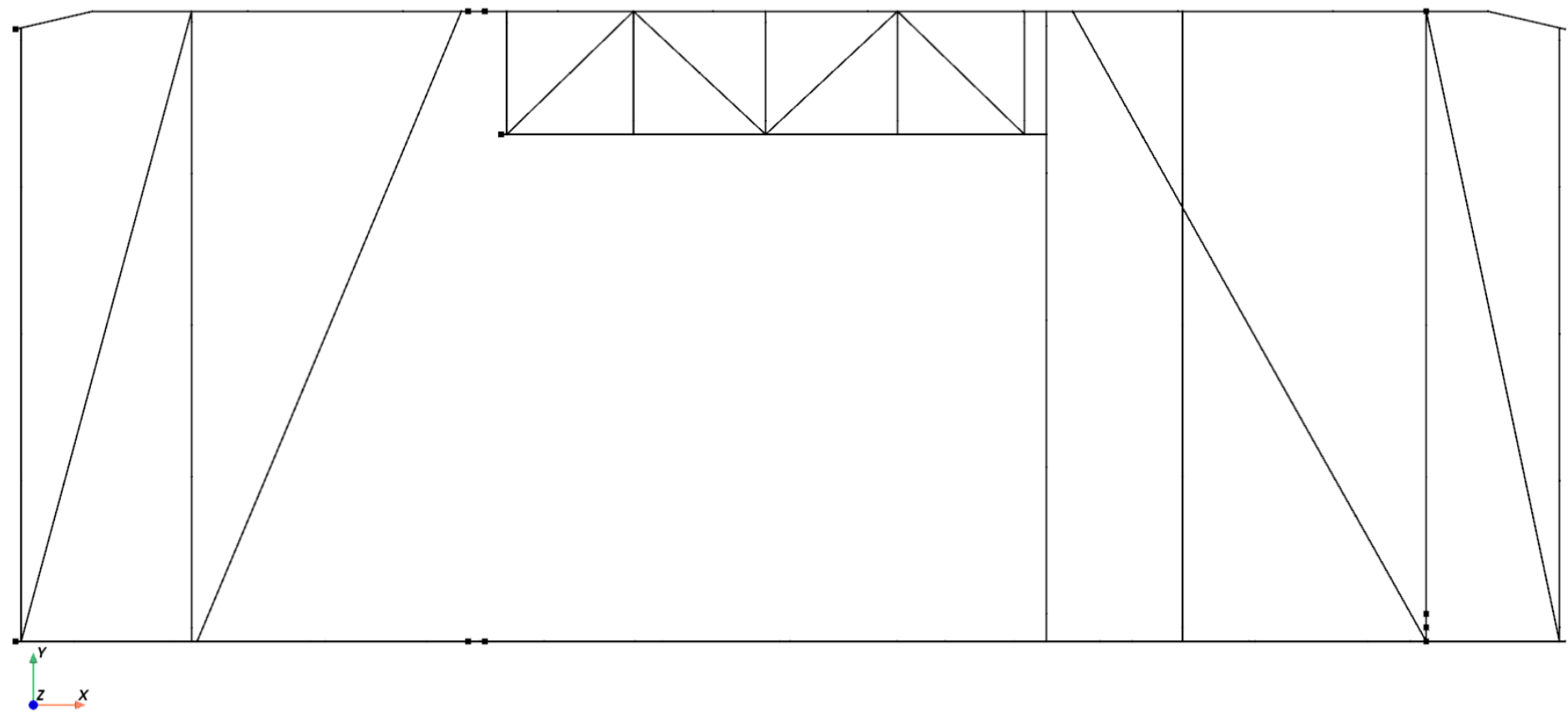
TN complete PyNite solved network

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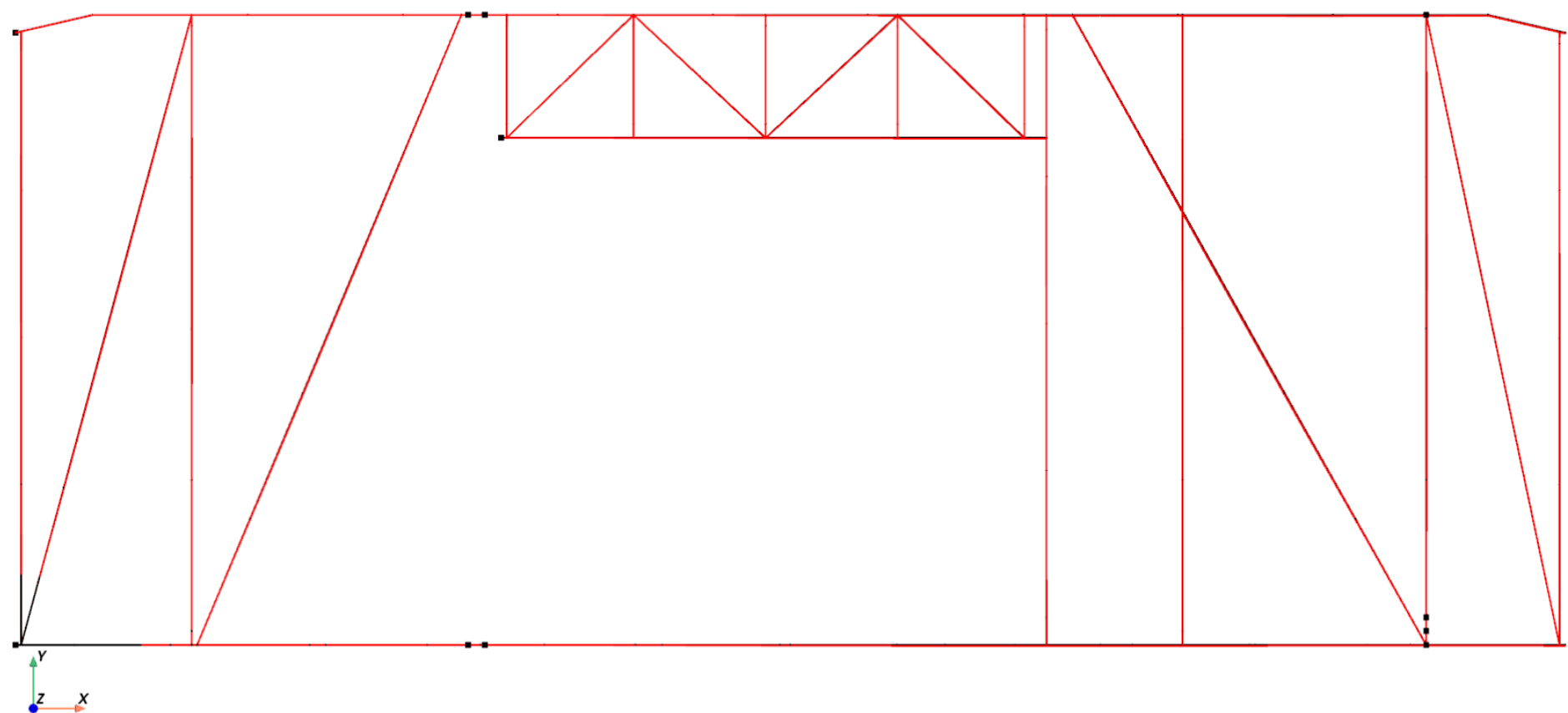
TN roof_first PyNite solved network



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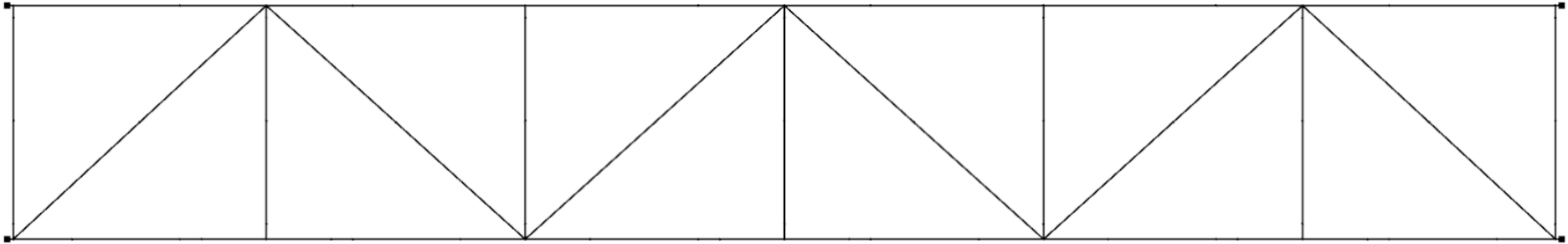
TN roof_first PyNite solved network

RED: displacement x20 roof_first_service



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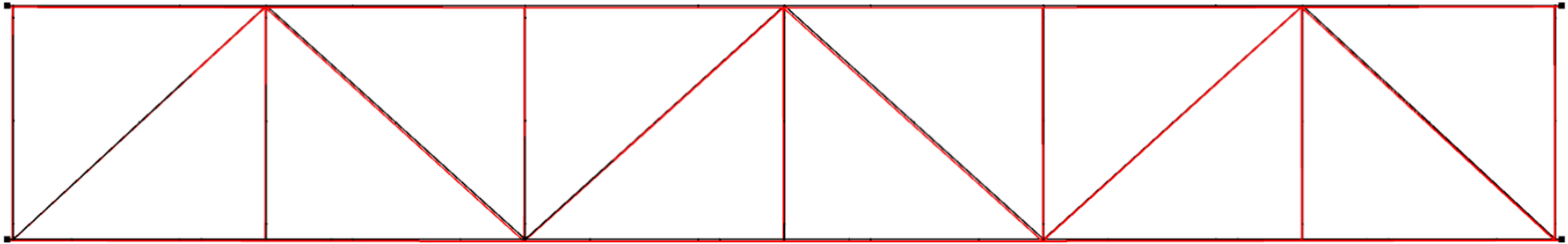
TS complete PyNite solved network



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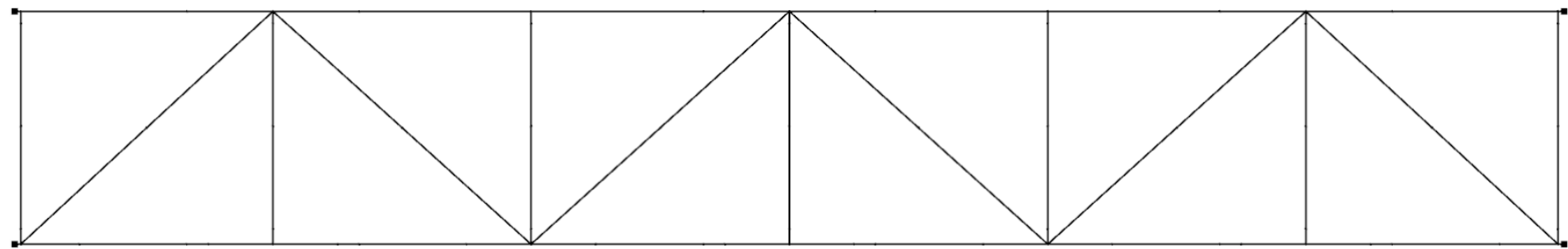
TS complete PyNite solved network

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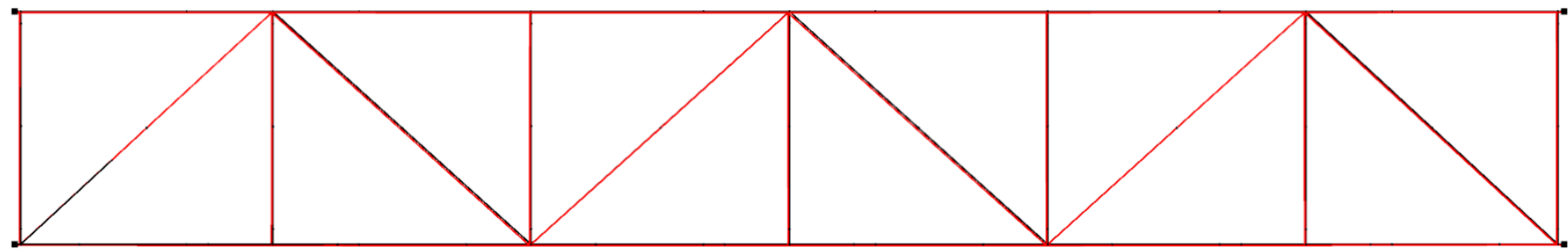
TS roof_first PyNite solved network



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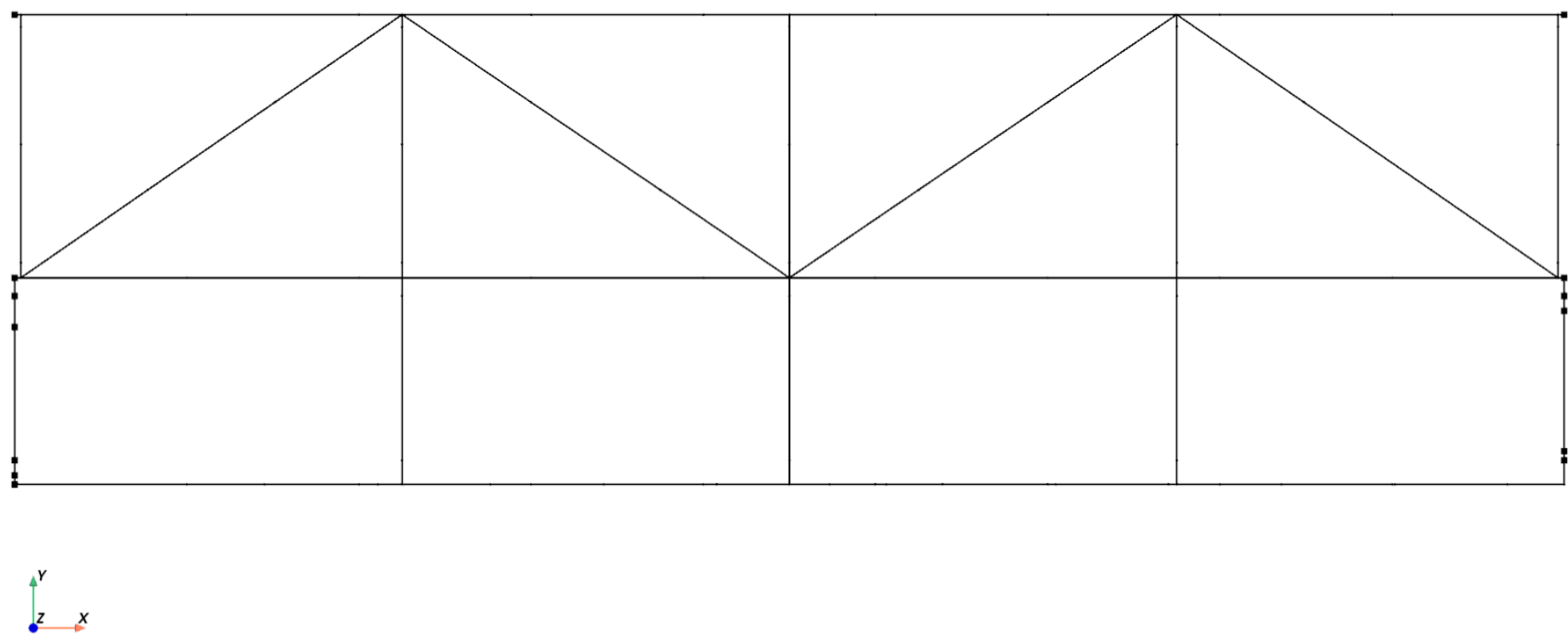
TS roof_first PyNite solved network

RED: displacement x20 roof_first_service



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Boundary glyphs represent imposed global-frame movements, not additional ground supports.

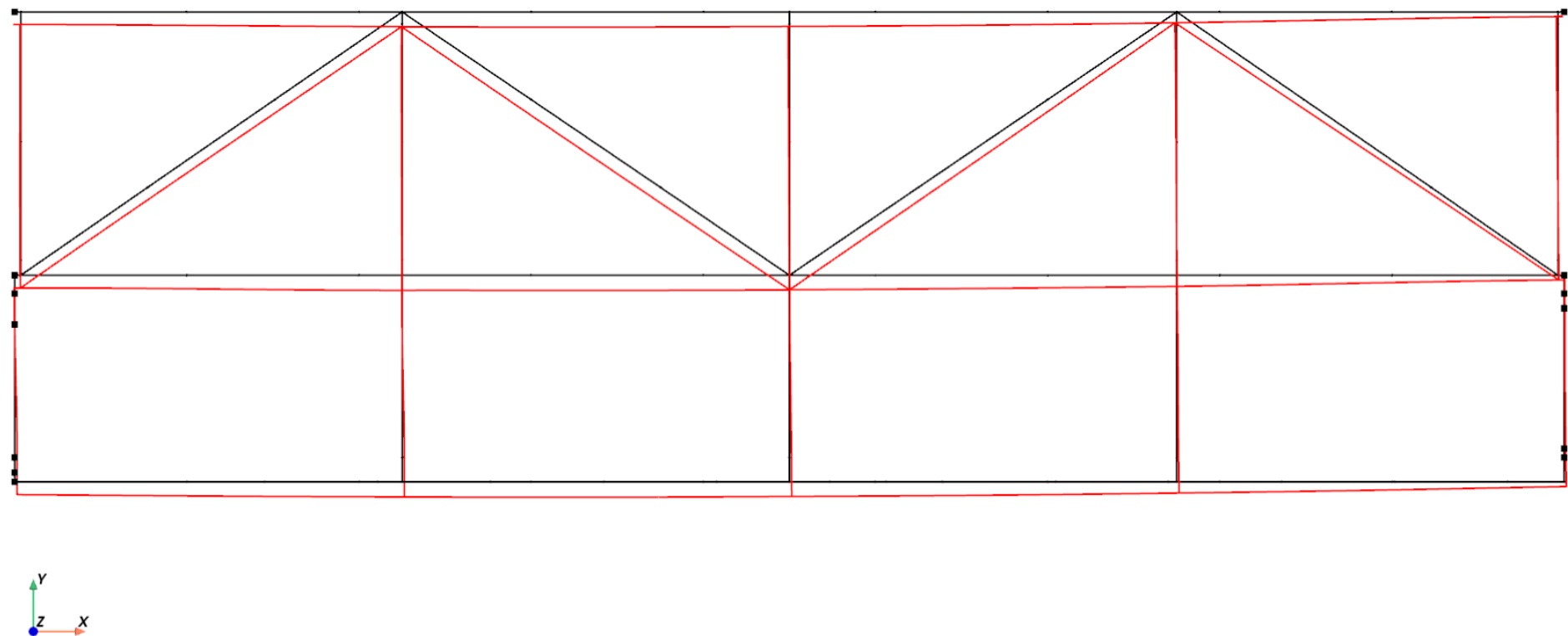
T1 complete PyNite solved network



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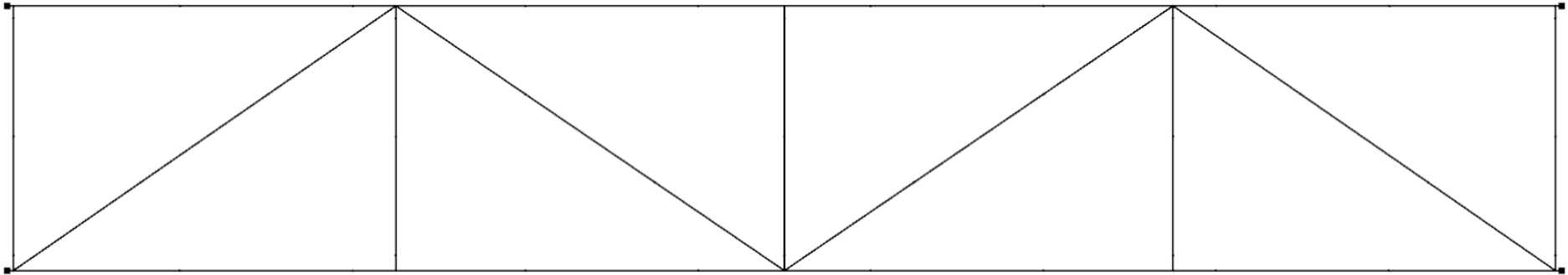
T1 complete PyNite solved network

RED: displacement x20 service_patches



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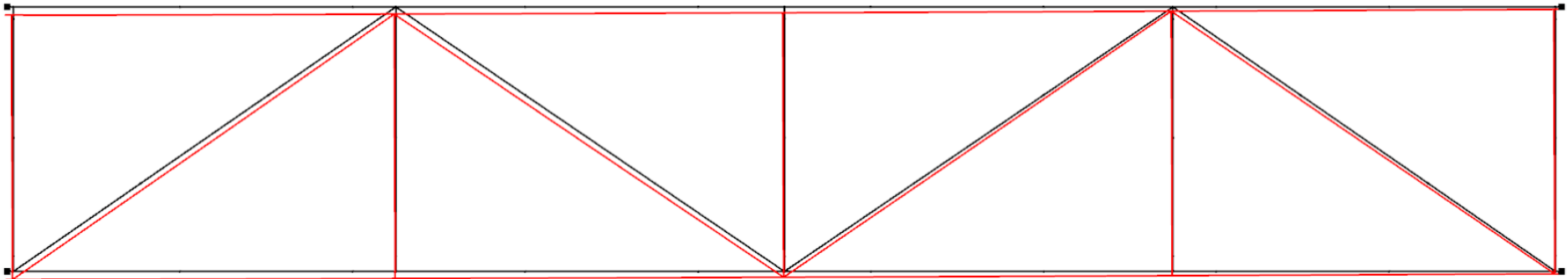
T1 roof_first PyNite solved network



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T1 roof_first PyNite solved network

RED: displacement x20 roof_first_service



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